

Real-Time Drowsiness Detection Based on Facial Features: A Rule-Based Method

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Abstract—Driver drowsiness is one of the leading causes of traffic accidents, especially during long-distance travel or nighttime driving. To address this issue, this study proposes a rule-based drowsiness detection system utilizing Eye Aspect Ratio (EAR), Mouth Aspect Ratio (MAR), and blink rate as a key indicators. Unlike complex machine learning-based systems, this approach does not require a large training dataset and focuses on predefined threshold rules for real-time detection. The proposed system is implemented using facial landmark detection via webcam and processes live video to identify signs of drowsiness. When the EAR value drops below a certain threshold, the MAR increases, or the blink rate exceeds a normal range, the system triggers an alert to notify the driver. Experimental results show that the rule-based system achieves an accuracy of 96.43%, which is lower than most machine learning-based methods. However, the simplicity of this approach makes it more efficient and easier to deploy on low-resource devices. This research highlights the trade-off between accuracy and system complexity, suggesting that rule-based detection can serve as a practical alternative in certain scenarios.

Keywords—Drowsiness Detection, Eye Aspect Ratio, Mouth Aspect Ratio, Blink Rate, Rule-based system

I. INTRODUCTION

Drowsiness while driving is a major contributor to traffic accidents worldwide. According to the World Health Organization (WHO), driver fatigue and drowsiness significantly increase the risk of road accidents[1]. As it is reported by NHTSA in March 2022, the number of fatalities involving drowsy drivers was 633 in 2020 [2]. Drowsiness impairs alertness, slows reaction time, and disrupts decision-making abilities, thereby escalating the likelihood of fatal incidents. This condition is particularly dangerous during long-distance travel or nighttime driving, where drivers are more prone to decreased vigilance. Therefore, detecting signs of drowsiness in real-time is essential for preventing accidents and saving lives.

Various methods have been developed to detect drowsiness, ranging from physiological monitoring (e.g., EEG, ECG) to behavioral and visual-based approaches[3]. Among these, visual behavior analysis is widely preferred due to its non-intrusive nature and ease of integration with standard camera devices. Several parameters have been proposed in this context, including Eye Aspect Ratio (EAR), Mouth Aspect Ratio (MAR), and blink rate, which are effective indicators of drowsiness. EAR measures eye closure duration, MAR reflects mouth opening intensity (often during yawning), and blink rate captures visual attention fluctuations, all of which are strong behavioral indicators of fatigue[4] [5].

Recent studies have focused on combining facial feature extraction with advanced machine learning algorithms to improve detection accuracy. For instance, one study extracted EAR, MAR, and head pose from facial landmarks and used classifiers such as random forest and support vector machines, achieving up to 99% accuracy on the National Tsing Hua University Drowsiness Detection dataset[3]. Another study implemented a hybrid method combining YOLOv3 for facial feature extraction and LSTM for modeling temporal behaviors such as blinking and yawning, showing robust performance even under illumination and head posture disturbances[6]. Meanwhile, the system called DriCare used 68 facial keypoints and a new face-tracking algorithm to detect fatigue signs like eye closure and yawning, reaching an accuracy of around 92%[7].

These approaches, although accurate, often require complex models, large datasets, and high-performance computing resources. In contrast, lightweight alternatives such as rule-based systems offer practical solutions for real-time applications in embedded systems or vehicles with limited computational capacity.

While machine learning (ML) and deep learning models have demonstrated high accuracy in drowsiness detection, they often require large datasets and high computational resources, making them less practical for real-time use in resource-constrained environments. Moreover, for simple embedded or vehicle-integrated systems, a lightweight yet reliable alternative is needed.

This research proposes a rule-based drowsiness detection system that uses EAR, MAR, and blink rate as key features. The system operates in real time, using a standard webcam to capture facial landmarks and trigger alerts when drowsiness is detected based on threshold values. Unlike ML-based approaches, this rule-based method does not require a training phase and can be easily deployed on devices with limited processing power.

The main contribution of this research lies in the development of a real-time, threshold-based drowsiness detection system that is simple and computationally efficient. Experimental results show that the system achieves a detection accuracy of 96.43%, which, although lower than typical ML-based systems, offers significant benefits in terms of speed, simplicity, and ease of implementation.

II. LITERATURE REVIEW

Research on driver drowsiness detection has advanced significantly in recent years, supported by the availability of specialized datasets such as the Real-Life Drowsiness Dataset (RLDD) and NTHU Drowsy Driver Detection (NTHU-DDD). These results highlight a clear trend: deep learning models such as LSTM and HM-LSTM consistently

outperform traditional classifiers in driver drowsiness detection tasks. Furthermore, the integration of facial landmarks and head pose estimation further improves the robustness of detection under various real-world conditions. While these methods achieve high accuracy, they often require considerable computational resources and labeled training data, which may limit their deployment in lightweight, real-time systems dataset. In 2019, Ghoddosian et al. utilized a Hierarchical Multiscale Long Short-Term Memory (HM-LSTM) model trained on the RLDD dataset and reported a model accuracy of 81%, outperforming human judgment which only reached 63% accuracy in identifying drowsiness[8].

In 2023, multiple studies using the NTHU-DDD dataset have shown continued improvements. Albadawi et al. compared several classifiers including Random Forest (RF), Sequential Neural Network, and Support Vector Machine (SVM). Their results showed that RF achieved 99% accuracy, followed by sequential NN with 96%, and SVM with 80%[3]. Similarly, Li et al. evaluated the performance of an LSTM network, which achieved the highest accuracy of 93.15%, compared to RF (83.49%) and SVM (85.19%) on the same dataset[5].

Safarov et al. proposed a face-mesh-based approach and found that the model was particularly effective in detecting drowsiness when the driver's head was down, achieving over 96% accuracy in those cases, and 85% in yawn-detection scenarios where both eye closure and mouth opening occurred[4].

Furthermore, a more recent approach proposed a Long-Term Multi-Granularity Deep Framework, which integrates a Multi-Granularity CNN (MCNN) for spatial feature extraction across various facial regions and an LSTM for temporal modeling. This method addressed challenges such as head pose variation and temporal dependencies and achieved 90.05% accuracy with a processing speed of 37 FPS on the NTHU-DDD dataset[9].

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III. RESEARCH METHODOLOGY

This section outlines the approach used to detect drowsiness by extracting visual features from facial regions, namely the eyes and mouth. The detection is based on three physiological indicators: Eye Aspect Ratio (EAR), Mouth Aspect Ratio (MAR), and blinking rate. The system is implemented using Python with OpenCV and a custom-trained dlib model. The system's process can be demonstrated in Figure 1.

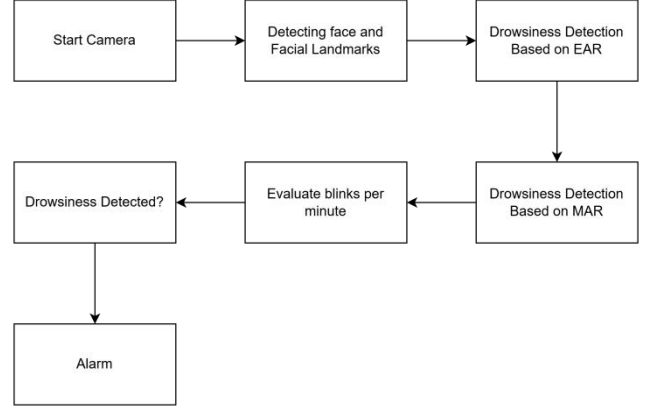


Fig. 1. Flow diagram of the system

The overall workflow of the proposed drowsiness detection system is illustrated in Fig. 1. The process begins by activating the camera, which continuously captures video frames of the driver's face. Each frame is then passed to a facial detection module that localizes the face and extracts facial landmarks using a custom-trained detector. Once the relevant eye and mouth landmarks are identified, the system computes the Eye Aspect Ratio (EAR) to assess eye openness. Simultaneously, the Mouth Aspect Ratio (MAR) is measured to detect signs of yawning. These features are individually compared against predefined threshold values to detect possible signs of drowsiness.

In addition to EAR and MAR, the system evaluates the blinking rate by analyzing eye closure patterns over a 60-second window. All three features—EAR, MAR, and blinking frequency—are then fed into a decision logic module that determines whether the driver is likely drowsy.

If drowsiness is detected based on any one of the three indicators, the system triggers an alarm to alert the driver in real time.

A. Dataset and Labeling

Two public video datasets were used: the UTAR Driver Drowsiness Dataset (UTARLDD) [8] and the Driver Drowsiness Detection Dataset (D3S) [10]. From these datasets, a subset of videos was selected to balance computational cost and maintain class representativeness. The data were grouped into two categories: drowsy and alert. The drowsy class includes frames where the subject is yawning, closing their eyes for prolonged periods, or exhibiting a blinking rate below the defined threshold.

Manual annotation was performed to refine the labels and to train a lightweight custom landmark detector for eyes and mouth. Instead of using the standard 68-point facial landmarks, a smaller set of key points was derived and used. The annotation process utilized the iBUG 300-W dataset to create a dedicated detector model using dlib's training utilities.

B. Preprocessing

Each video frame was converted to grayscale to reduce computational load and improve robustness. Additionally, frames were resized to a uniform size. However, no further normalization was applied to standardize varying resolutions across different videos due to current system limitations.

C. Feature Extraction

Facial feature extraction was performed using a custom-trained dlib shape predictor, optimized to detect only the eye and mouth regions. The default 68-point facial landmark model was replaced with a lightweight model trained on a modified version of the iBUG 300-W dataset, annotated specifically to include only the relevant landmarks for eyes and mouth.

This custom predictor was trained using dlib's supervised descent algorithm, resulting in faster and more focused detection. After landmark localization, three visual features were extracted:

- Eye Aspect Ratio (EAR) – Measures eye openness by computing vertical-to-horizontal eye landmark ratios.
- Mouth Aspect Ratio (MAR) – Captures mouth openness, often associated with yawning.
- Blinking Rate – Estimated by counting the number of full eye closures per minute.

The mathematical definitions for EAR and MAR are given below:

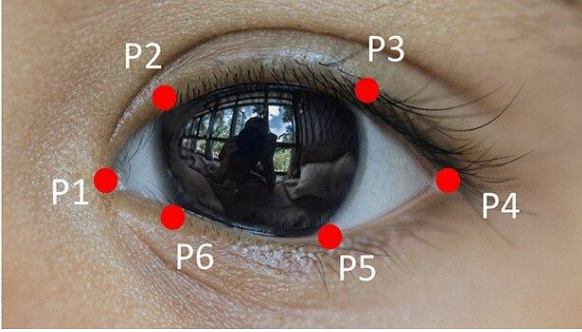


Fig. 2. Eye landmarks point [11]

$$EAR = \frac{||p2 - p6|| + ||p3 - p5||}{2||p1 - p4||} \quad (1)$$

Where p1 and p4 are the horizontal eye corner points, p2 and p6, p3 and p5 are vertical pairs of eye landmarks. $||$ denotes the Euclidean distance between two points.

A decrease in EAR over a sustained period indicates a potential eye closure, which is a strong indicator of drowsiness. When the EAR falls below a certain threshold (0.2 in this study), the system flags the frame as representing a drowsy state. This threshold was selected based on experimental validation from [3]. Furthermore, if the EAR value drops to ≤ 0.1 in a single frame, the system categorizes it as a blink, which is then used to calculate the subject's blinking rate as an additional feature for drowsiness detection.

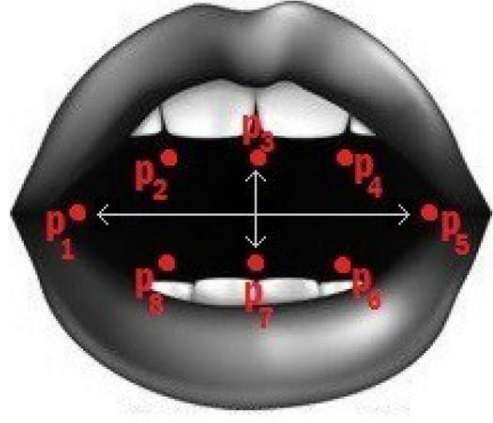


Fig. 3. Mouth landmarks point [3]

$$MAR = \frac{||p2 - p8|| + ||p3 - p7|| + ||p4 - p6||}{2||p1 - p5||} \quad (2)$$

Where p1 and p5 are the left and right corners of the mouth. P2 to p4 and p6 to p8 represent vertical pairs of landmarks along the inner lip contour. $||$ denotes the Euclidean distance between two points.

When the mouth is opened widely—as occurs during yawning—the vertical distances increase, leading to a higher MAR value. A MAR exceeding the threshold of 0.5 is used in this study to indicate the presence of a yawn, based on validated thresholds reported in [3].

D. Drowsiness Detection Logic

A rule-based approach was adopted to determine the drowsiness state. Threshold values for EAR and MAR were selected based on the results from [3], where all data frames from all subjects were successfully labeled as either “open eyes” or “closed eyes,” and similarly for “open mouth” or “closed mouth.” From this study, an EAR threshold of 0.2 and a MAR threshold of 0.5 were selected as the optimal boundaries for drowsiness classification.

For blinking rate, reference values were taken from [12], which state that drowsy individuals typically blink only 4–6 times per minute, compared to approximately 10 times for alert individuals. To identify blink events at the frame level, the system uses a stricter EAR threshold drops below 0.1 to capture momentary closures of the eyes, which represent blinks.

In this system, a video is classified as drowsy if any one of the following conditions is met:

- The EAR drops below 0.2 for a sustained number of frames, indicating prolonged eye closure;
- The MAR exceeds 0.5, suggesting yawning behavior;
- The number of detected blink events—defined as instances where the EAR drops to ≤ 0.1 in a single frame—falls below 6 blinks per minute, indicating a reduced blinking frequency commonly associated with fatigue.

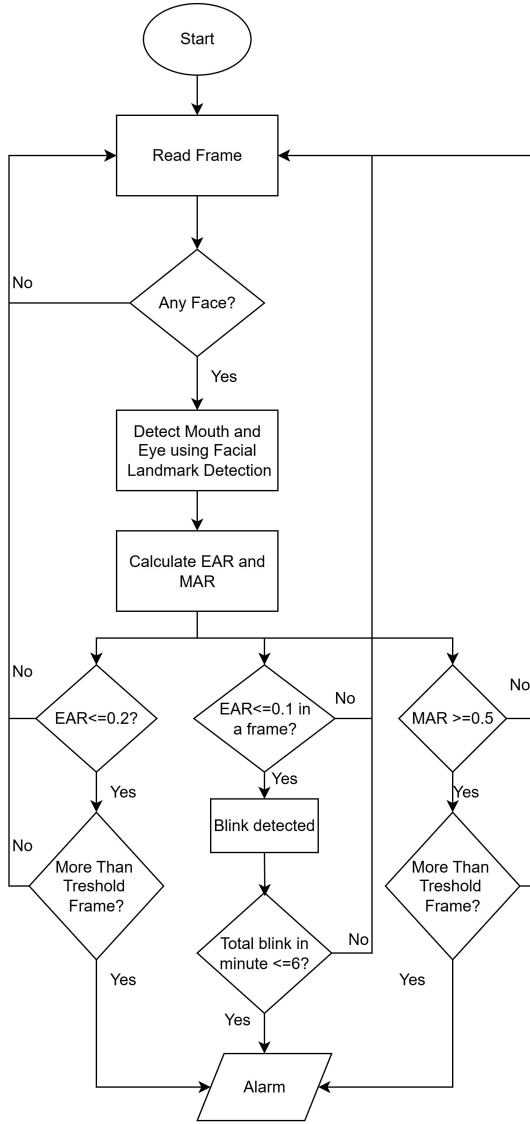


Fig. 4. Flowchart The System

E. Evaluation

The system's performance was evaluated using accuracy F1-score as the primary metric. An ablation study was performed to assess the contribution of each feature. The following feature combinations were tested:

- EAR only
- MAR only
- EAR + MAR
- EAR + Blinking Rate
- MAR + Blinking Rate
- EAR + MAR + Blinking Rate

This analysis highlights the importance of combining multiple physiological indicators for more reliable drowsiness detection.

IV. RESULT AND DISCUSSION

The proposed drowsiness detection system was evaluated using a total of 28 video samples, consisting of 11 alert and 17 drowsy instances. These samples were drawn from both

the D3S and UTARLDD datasets. All videos were used exclusively for testing, as the system does not rely on a training phase due to its rule-based nature.

To assess the effectiveness of each feature in detecting drowsiness, a series of ablation experiments were conducted. The features evaluated include Eye Aspect Ratio (EAR), Mouth Aspect Ratio (MAR), and blinking rate—both individually and in various combinations. Accuracy was used as the primary evaluation metric, calculated as the percentage of correctly classified videos in the test set.

A. Evaluation Results

Table I presents the accuracy achieved under different feature configurations. The EAR feature alone provides a solid baseline, as prolonged eye closure is a strong drowsiness indicator. Similarly, MAR alone reflects yawning behavior, which is also common during drowsy states. The inclusion of blinking rate adds temporal context to the facial behavior and helps capture subtle signs of fatigue.

TABLE I. COMPARISON FEATURE DROWSINESS DETECTION

Combination	Accuracy
EAR only	75%
MAR only Feature	53.57%
Blinking Rate only	57.14%
EAR + MAR	82.14%
EAR + Blinking Rate	89.29%
MAR + Blinking Rate	71.43%
EAR + MAR + Blinking Rate	96.43%

To contextualize the performance of the proposed rule-based system, a comparison was made with the results reported by Albadawi et al. [3], who evaluated three different machine learning approaches for driver drowsiness detection: Linear Support Vector Machine (SVM), Random Forest (RF), and a Sequential Neural Network. Their experiments achieved accuracies of 80%, 99%, and 96%, respectively. In contrast, the proposed rule-based system achieved an accuracy of 96.43% using only handcrafted visual features (EAR, MAR, and blinking rate) and thresholding, without requiring any training process. While the Random Forest model in their study slightly outperformed the rule-based system in terms of accuracy, the proposed approach offers significant advantages in terms of simplicity, transparency, and suitability for real-time and low-resource environments. This comparison suggests that, under appropriate conditions, rule-based systems can remain competitive with more complex learning-based models, particularly when interpretability and efficiency are prioritized.

TABLE II. COMPARISON WITH SOME METHODS

Method	Feature	Approach	Accuracy
Albadawi et al. – Linear SVM [3]	Eye, Mouth, and Head	SVM	80%
Albadawi et al. – RF [3]	Eye, Mouth, and Head	Random Forest	99%
Albadawi et al. – NN[3]	Eye, Mouth, and Head	Seq. Neural Net	96%
This study	Eye, Mouth, and Blinking Rate	Rule Based	96.43%

The experimental results, summarized in Table I, show that the Eye Aspect Ratio (EAR) alone achieved an accuracy of 75%, while the Mouth Aspect Ratio (MAR) and blinking rate alone achieved lower accuracies of 53.57% and 57.14%, respectively. Combining features improved performance significantly. The EAR and MAR combination achieved 82.14%, and EAR with blinking rate reached 89.29%. The highest accuracy, 96.43%, was obtained when all three features were combined, demonstrating the effectiveness of multi-feature fusion in enhancing drowsiness detection.

Notably, the rule-based system demonstrates promising performance without the need for a machine learning model or training process. This highlights its potential for real-time deployment in low-resource environments, such as embedded systems in vehicles or mobile applications.

Despite its simplicity, the system has several advantages:

- Lightweight and interpretable : Each decision is based on well-understood physiological rules.
- No need for annotated training data: Eliminates the cost and complexity of supervised learning.
- Real-time capable: Fast processing due to limited landmark detection and absence of inference time for classifiers.

However, some limitations remain. The system's performance may degrade under varying lighting conditions, occlusions (e.g., glasses or hands), or in cases where users suppress typical drowsy behavior. Additionally, the absence of adaptive thresholds may reduce generalizability across diverse subjects and environments. The system is also sensitive to head pose variation, where significant tilting or rotation of the face relative to the camera may hinder accurate landmark detection and feature computation.

Future improvements may include standardizing input video resolutions, incorporating adaptive or subject-specific thresholds, or combining the rule-based system with lightweight machine learning models to enhance robustness and flexibility.

V. CONCLUSION

This paper presented a rule-based driver drowsiness detection system that utilizes three visual features: Eye Aspect Ratio (EAR), Mouth Aspect Ratio (MAR), and blinking rate. The system was designed for simplicity, interpretability, and real-time applicability, without requiring any machine learning model or training phase.

Using video samples collected from the D3S and UTARLDD datasets, the system was evaluated through a series of ablation experiments. The results demonstrated that while individual features provide useful indicators of drowsiness, combining multiple features significantly improves detection performance. The highest accuracy was achieved when EAR, MAR, and blinking rate were used together, suggesting that integrating complementary facial cues enhances the system's robustness.

The system offers several practical advantages, including ease of deployment, fast processing, and independence from annotated training data. However, limitations remain, such as sensitivity to lighting conditions, occlusions (e.g., glasses, hands), head pose variations, and the use of fixed thresholds.

These issues may affect robustness and generalizability across subjects and environments.

Future work will focus on addressing these limitations by introducing adaptive or subject-specific thresholds, incorporating head pose correction techniques, and evaluating the system in real-world driving scenarios. Additionally, a comparative study with lightweight machine learning-based approaches may further strengthen the system's relevance and performance.

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